Tairan Song

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Education

Michigan State University, East Lansing, MI, USA
Ph.D. Student in Computational Mathematics, Science and Engineering; **GPA: 3.8 / 4.0** Aug 2023 – Present
M.S. in Computational Mathematics, Science and Engineering; **GPA: 3.8 / 4.0** Aug 2023 – Aug 2026
Advisor: Dr. Adam Alessio

King's College London, London, Greater London, United Kingdom
M.Sc. in Applied Statistical Modelling and Health Informatics; **Graduated with Distinction** Sep 2021 – Sep 2022
Advisors: Dr. Ioannis Bakolis and Dr. Amy Ronaldson

University of Liverpool, Liverpool, Merseyside, United Kingdom
B.Sc. in Mathematics and Statistics; **Upper Second-Class Honours** Sep 2018 – Jun 2021
Advisor: Dr. Jinlai Li

Publications

- **Song, T.**, Huang, Z., Noyes, S. L., et al., & Alessio, A. (2026). Supervoxel Graph Representation Learning for Renal Mass Classification from CT. Submitted to **SPIE Medical Imaging 2027**. **Under Review**.
- Munavar Ali, M. A., **Song, T.**, Noyes, S. L., et al., & Alessio, A. (2026). Histologic Grade Prediction in High-Risk Renal Masses from Multi-Region CT Analysis. Submitted to the **2026 Leaders on the Horizon Residents' Program**, **Applied Radiology**. **Under Review**.
- Zhang, T., Jia, H., **Song, T.**, Lv, L., Gulhan, D. C., Wang, H., … & Shen, N. (2023). De novo identification of expressed cancer somatic mutations from single-cell RNA sequencing data. **Genome Medicine**, 15(1), 115. doi: 10.1186/s13073-023-01269-1

Conference Presentations & Abstracts

- **Song, T.**, Noyes, S. L., Munavar Ali, M. A., et al., & Alessio, A. (2026). Core-to-Margin Radiomic Profiling for Predicting Kidney Cancer Grade from Variable-Phase CT. Accepted scientific poster presentation at the **Radiological Society of North America (RSNA) 2026 Annual Meeting**, Chicago, IL, Nov 29–Dec 3, 2026.
- Muterspaugh, R., Munavar Ali, M. A., **Song, T.**, et al. (2026). Deep Learning-Based Detection of High-Risk Renal Masses from All-Cause Abdominal CT. Oral presentation at **Corewell Health West Research Day**, Grand Rapids, MI, May 8, 2026. Abstract 1968.
- **Song, T.** (2024). Enhancing Transcription Factor Binding Site Prediction with Epigenetic Data: A BERT-Based Machine Learning Innovation. Poster presented at the **20th Annual Meeting of the MidSouth Computational Biology and Bioinformatics Society (MCBIOS)**, Emory University, Atlanta, GA, Mar 22–24, 2024.

Research Experience

Spatial Graph Modeling of Renal Tumor Heterogeneity from Abdominal CT
Advisor: Dr. Adam Alessio Michigan State University Jan 2026 – Present

- Developing supervoxel graph representations of renal masses from abdominal CT to characterize intratumoral spatial heterogeneity and core-to-margin organization, with graph-based and transformer-based representation learning for renal mass classification.
- Investigating spatially informed feature representations that integrate local image characteristics with graph topology to capture heterogeneous tissue organization beyond conventional whole-tumor radiomic summaries.

Quantitative CT Analysis for Renal Mass Characterization under Variable Contrast Timing
Advisor: Dr. Adam Alessio Michigan State University Jan 2025 – Dec 2025

- Developed an end-to-end quantitative imaging pipeline for renal mass characterization from heterogeneous abdominal CT examinations with variable contrast timing, incorporating anatomical segmentation with TotalSegmentator, renal mass segmentation with a custom nnU-Net, and radiomic feature extraction.
- Developed multi-region feature representations combining quantitative descriptors from the renal mass with reference anatomy, including normal kidney, aorta, and inferior vena cava, to account for variability in contrast enhancement across CT examinations.
- Evaluated radiomic, spatial, and reference-region feature combinations with supervised machine-learning classifiers under cross-validation, demonstrating improved renal mass classification performance compared with tumor-only feature representations.

De Novo Identification of Expressed Cancer Somatic Mutations from scRNA Sequencing Data
Advisor: Dr. Ning Shen Zhejiang University Sep 2022 – Mar 2023

- Contributed to the development and evaluation of RESA (Recurrently Expressed SNV Analysis), a computational framework for de novo identification of expressed somatic mutations from scRNA-seq data.
- Implemented and validated the joint logistic regression component (RESA-jLR), and performed benchmark analyses and visualization to evaluate mutation-calling performance against existing methods.

Teaching Experience

Teaching Assistant — Michigan State University

- CMSE 202: Computational Modeling and Data Analysis II — Jan 2025 – May 2025

Internship Experience

Zhejiang University — Hangzhou, China
Research Assistant — Sep 2022 – Mar 2023

China Galaxy Securities Co., Ltd. — Shanghai, China
Financial Data Analytics Intern — Jul 2021 – Sep 2021

Awards and Honors

NSF NRT-IMPACTS Training Support — Michigan State University — Fall 2023
Received one-semester graduate training support through the NSF Research Traineeship-funded IMPACTS Program (NSF DGE-1828149).

Skills

Programming: Python, R, SQL
ML & AI: PyTorch, scikit-learn; deep learning for medical imaging (CNN-based architectures, e.g., U-Net); NLP and Transformer-based models
MLOps & Systems: SLURM, Git, High-Performance Computing
Research: Quantitative Medical Imaging, CT Image Analysis, Radiomics, Medical Image Segmentation, Graph Representation Learning, Biomedical Data Analysis, Deep Learning